

CSC 461/592 – Assignment #1 Unity Basics

Goal

Familiarize yourself with the Unity Engine and its capabilities for basic 3D environment design.

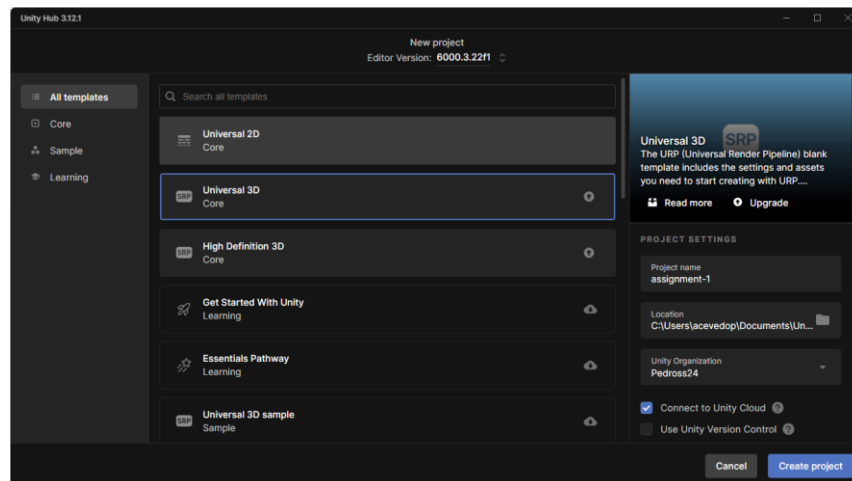
Description

You will create three Unity scenes that demonstrate your ability to build 3D environments, use Unity assets, and set up a VR-ready scene. The focus is on practicing fundamentals: primitives, components, and scene design.

Requirements

Setup

- Use Unity **6000.3.22f1LTS** (download from the [Unity Archive](#)).
- Create a new project, make sure to select the “Universal 3D” project.



- In your Unity project, create an “*Assignments*” folder inside the “*Assets*” folder.
- Inside it, create another folder called “*Unity Basics*” where you will store all assets and scenes.
- Create a GitHub repository for this project and ensure it has the correct Unity .gitignore and settings (see here [on Medium](#)).

Task

- **Scene 1 – Unity Primitives:** create a scene with at least 3 different shapes and 3 different materials.
- **Scene 2 – Maze with Character Controller:** Use a Character Controller ([Unity Asset Store](#)) to allow navigation through a maze. The camera must be used in a

top-down perspective. (If you import the package, also add the *Cinemachine* package.)

- **Scene 3 – VR 360° Scene:** Load a 360° image or Skybox viewable in a VR headset (see [Unity Docs](#)).

Bonus (Optional - at least one mandatory for master's students)

- **BONUS 1:** Create a script to change the color of a material at runtime (see [YouTube](#)).
- **BONUS 2:** Load a 360 video and display it in VR (see [Unity Learn](#)).
- **BONUS 3:** Make the maze also accessible through the VR HMD (see [Meta](#)).

Submission

- Submit a GitHub repository link containing your full Unity project.
- The project must include the three required scenes, working for Unity 6000.3.22f1 LTS.
- In your repository README.md, include:
 - A short description of each scene.
 - At least one screenshot for each scene.
 - If you implemented a bonus, label it clearly (e.g., Scene 2 – Maze (Bonus 3: VR HMD Accessible)).

Example README format:

<pre># CSC 461/592 – Assignment 1 Unity Basics ## Scene 1 – Unity Primitives This is a scene with... (image-link) ## Scene 2 – Maze This maze has one corridor... (image-link) ## Scene 3 – VR 360 There is a 360° view of a... (image-link)</pre>	<pre>CSC 461/592 – Assignment 1 Unity Basics Scene 1 – Unity Primitives This is a scene with... Screenshot1 Scene 2 – Maze This maze has one corridor... Screenshot2 Scene 3 – VR 360 There is a 360° view of a... Screenshot3</pre>
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Notes:

- The last commit before the deadline will count as your submission. Changes to the assignment after the deadline will not be reviewed.
- Master's students must complete at least one bonus feature.

Resources

- GitHub with Unity ([Medium](#))
- Adding an image to a README ([Stack Overflow](#))

- Character Controller example ([YouTube](#))
- Adding a Skybox ([Unity Docs](#))